Week 3: Possible Direction

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American option valuation can be formulated as a free-boundary problem, where the optimal early-exercise boundary is unknown and must be determined jointly with the price function. This introduces two main numerical challenges.

1. Terminal kink: The solution is non-smooth near the strike price due to the kink in the payoff function. This lack of smoothness propagates backward in time and makes numerical methods harder to stabilize and train.

2. Free boundary: The early-exercise boundary is unknown in advance and must be determined as part of the solution, while ensuring the constraint $V(S,t) \geq K - S$ is satisfied.

These regions exhibit strong nonlinearity and localized features, suggesting that uniform discretization or sampling may be inefficient. Recent work indicates that neural solvers may struggle to accurately capture these regions without specialized treatment. To solve the difficulties, possible directions with relevant references:

1. A general neural PDE solver for option pricing: [6] shows feasibility and benchmarking against classical numerical methods rather than solving the hardest American-option-specific issues.

2. Algorithmic Directions Improvement:

   - Structure-aware: [7] concerns the treatment of terminal conditions. The payoff function is inherently non-smooth, which leads to instability near maturity. They propose enforcing exact terminal conditions and designing network structures that better match the singular behaviour near expiration.
   - More training steps: [3] proposes the algorithmic trick to perform more training steps of the solution network against a single step of the free boundary at each epoch.
   - Adaptive sampling: Limited direct practice, direct example in [2].

3. Financial Perspective Evaluation: Most existing studies compare results with analytical or numerical benchmark solutions. However, a more realistic evaluation would involve comparison with observed market option

prices. (e.g., [5]–Data driven method, [1]–Data source and evaluation) And including more evaluation on risk management quality: from the outcome of stability measured by Delta and Gamma, to ask whether smoother neural estimates lead to better hedging or more stable scenario analysis under volatile market regimes.

4. Beyond BS equation: Asset prices may exhibit jumps or regime-switching behaviour, leading to partial integro-differential equations (PIDEs) instead of classical PDEs [4].

References

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